

a Encode duplex

Sequence + chemistry
Guide / passenger



$X : 64 \times 93$
 $A : 8 \times 64 \times 64$
Padding mask

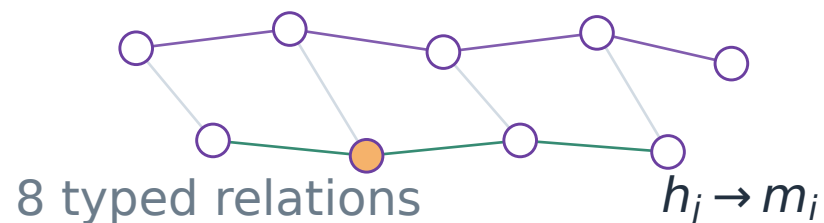
Training-only masks
Features + context

8 context fields

b Typed message passing

2 layers · hidden width 32

Linear $93 \rightarrow 32 + \text{ReLU}$
 $H^0 : 64 \times 32$



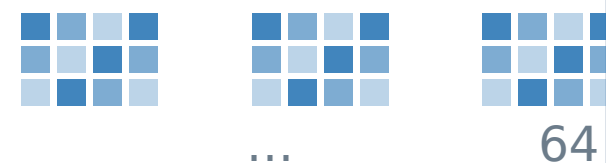
$$m_i = d_i^{-1} \sum_{r,j} A_{rij} h_j \widetilde{W}_r$$
$$d_i = \max(1, \text{total degree}_{r,j})$$

Self transform + message
Residual + norm + mask

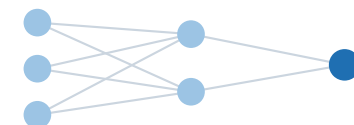
R1: shared + gated routes

c Ordered readout

Flatten H^2 in order



2,048 coordinates
+ 8 context fields



Dense 2,056 $\rightarrow$ 64
ReLU + dropout 0.1

Linear $64 \rightarrow 1 : z$

$$\hat{y} = \mu_T + S_T Z$$

d Evaluate

Freeze weights
Disable dropout

B1 · Activity / ranking
One condition

B2 · Pair effect
 $\Delta y = y_1 - y_0$
Fixed sequence + assay

B3 · Interaction
 $00^+ \quad 10^-$
 $01^- \quad 11^+$
Four measured conditions
Shared controls
 $y_{11} - y_{10} - y_{01} + y_{00}$

Activity training

FIT

Activities

$$\tilde{y} = (y - \mu_T) / S_T$$

Weighted MSE

$$(z - \tilde{y})^2$$

Update

shared weights

Measured training labels

Pair-supervised follow-up

FIT

Pairs

$$\Delta y = y_1 - y_0$$

Activity loss + λ

weighted pair loss

Update

shared weights

Held-out labels: evaluation only

e MODEL AUDIT: PRE-REGISTERED INTERVENTION FITS ONLY

siRNA mask restoration: 56 fits / MAVE-NN: 6 fits · 12 attempts

Saved predictions
+ saved checkpoints

Exact evaluation IDs
Row / study weighting

Source decomposition
V, MSE and covariance

Complete-input classes
 $\text{rank}(CH) = 72$
140 contrast rows

Bounded perturbations
Training outputs fixed

Published-predictor evaluation
Matched protocols and weights